

Lab 3: Component Modelling & Architectural Pattern Selection

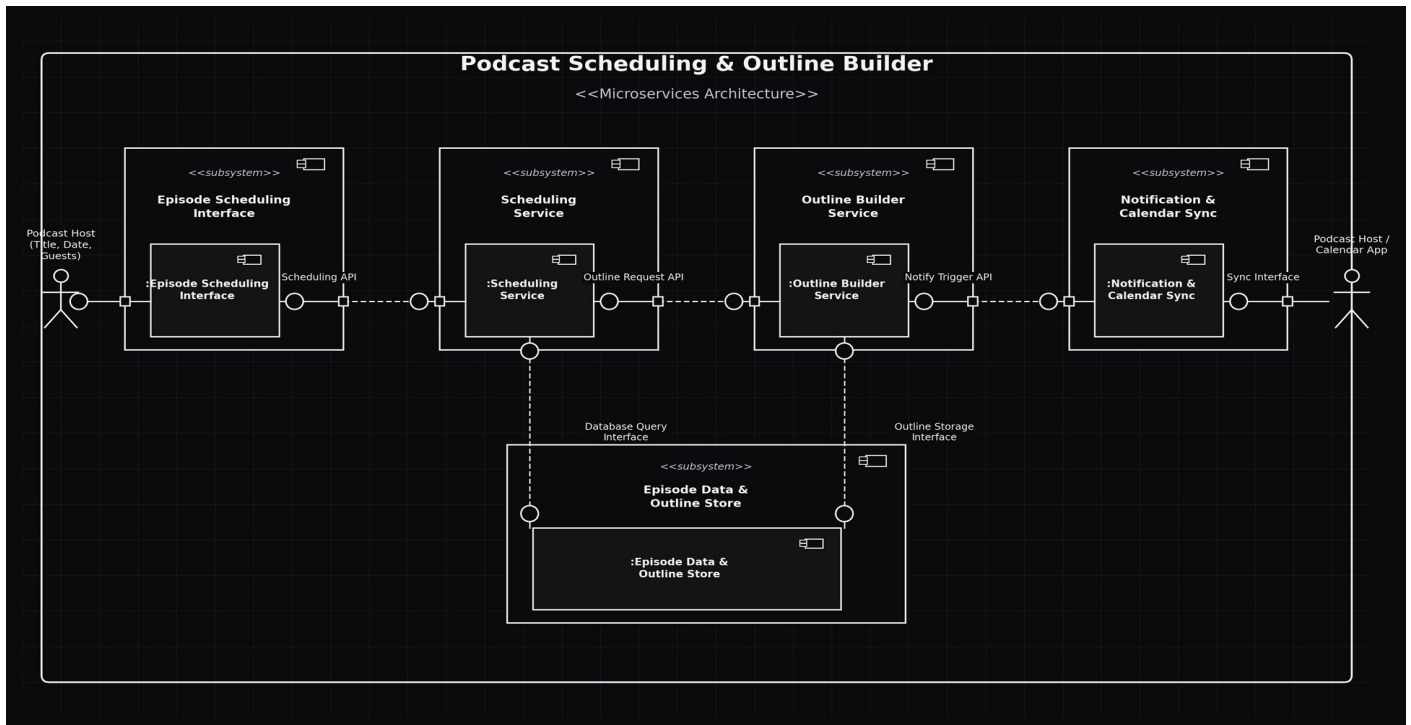
Podcast Scheduling & Outline Builder

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Deliverable 1: Component Diagram

Podcast Scheduling & Outline Builder — Microservices Architecture



Deliverable 2: Written Justification

System: Podcast Scheduling & Outline Builder

Architecture Selection: We chose Microservices Architecture for the Podcast Scheduling and Outline Builder System.

Architectural Choice

The system is decomposed into independent services — Episode Scheduling Interface, Scheduling Service, Outline Builder Service, Notification & Calendar Sync Service, and an Episode Data & Outline Store — each communicating through well-defined REST/API interfaces rather than shared code or a single monolithic codebase.

Reason 1: Independent Scaling for Variable Load

AI-driven outline generation is computationally heavy and used in short, unpredictable bursts (e.g., right before a recording session), while scheduling and calendar sync see steady, lightweight traffic. Microservices let the Outline Builder Service scale independently on demand without over-provisioning the rest of the system, which would be impossible in a single-tiered client-server deployment.

Reason 2: Independent Deployment and Technology Fit

The Outline Builder relies on an AI/NLP model stack that evolves rapidly, while Scheduling and Calendar Sync are stable, business-rule-driven services. Microservices allow each team to update, redeploy, or even reimplement the Outline Builder in a different language or model version without redeploying or risking downtime in the Scheduling or Notification services.

Security Advantage

Fault and access isolation: the Episode Scheduling Interface is the only component directly exposed to podcast hosts, so a compromise or bug in the Outline Builder or Calendar Sync service cannot directly expose user credentials or scheduling data — each service is independently sandboxed and only reachable through its declared interface.

Performance Benefit

Because outline generation runs asynchronously behind its own API, slow AI response times do not block the Scheduling Service from remaining responsive for calendar sync or reminder notifications, improving perceived performance and overall system responsiveness for users.